

Lecture 4 - Exercises

- Exercise 1:

Write pseudocode for Lamport clock algorithm in the API style from lecture 1 (see slide example for failure detector for formatting ideas).

- Exercise 2:

Calculate the Lamport clocks for the Lamport diagram in slide 9. Explain if they can be used to fix the issue presented in that slide.

- Exercise 3:

Prove that If $a \rightarrow b$ then $LT(a) < LT(b)$.

- Exercise 4:

Assume a set of nodes, and two events occurring in two different nodes with $LT(e_1) = 5$ and $LT(e_2) = 3$. What can you say about the causality of events e_1 and e_2 ?

- Exercise 5:

In slide 13 list all possible FIFO broadcast allowed orders of the messages a, b, and c. In slide 14 list all possible causal broadcast allowed orders of the messages a, b, and c.

- Exercise 6:

Prove that all Causal broadcast protocols are also FIFO broadcast protocols.

- Exercise 7:

Write pseudocode for a causal broadcast. You will be needed to keep track of some vector clock structure.

- Exercise 8:

Can we devise a uniform reliable broadcast algorithm with an eventually perfect failure detector but without assuming a majority of correct processes?

- Exercise 9:

Compare the causal delivery property with the following property: "If a process delivers messages m_1 and m_2 , and $m_1 \rightarrow m_2$, then the process must deliver m_1 before m_2 ."

- Exercise 10:

Calculate the vector clocks of the following distributed system. Show the contents of two different exchanged messages. Give an example of 2 different pairs of concurrent events.

